

Functional Annotation Report: MurB (Q88LM5, PP_1904) — *Pseudomonas putida* KT2440

1. Summary / Answer

MurB (UniProt Q88LM5; gene *murB*; ordered locus PP_1904) of *Pseudomonas putida* KT2440 is UDP-N-acetylenolpyruvoylglucosamine reductase (EC 1.3.1.98; also called UDP-N-acetylmuramate dehydrogenase). It is a cytoplasmic, monomeric FAD-dependent flavoenzyme that catalyzes the second committed step of the cytoplasmic phase of peptidoglycan (cell-wall) biosynthesis: the NADPH-dependent reduction of the enolpyruvyl group of UDP-N-acetylglucosamine-enolpyruvate (the product of MurA) to yield UDP-N-acetylmuramic acid (UDP-MurNAc), the essential precursor that MurC–MurF subsequently convert into UDP-MurNAc-pentapeptide. The enzyme is essential for viability and is a validated antibacterial drug target.

Gene-identity verification

The gene symbol **murB** matches the UniProt protein description (UDP-N-acetylenolpyruvoylglucosamine reductase, EC 1.3.1.98, MurB family), the organism (*Pseudomonas putida* KT2440), and the domain architecture (FAD-bd_PCMH cofactor-binding fold; MurB signature IPR003170). The literature retrieved is unambiguously for this enzyme family. No mechanistic study exists on the *P. putida* KT2440 protein itself; however, the enzyme is one of the most thoroughly characterized in bacterial cell-wall biology, and — importantly — a high-resolution crystal structure exists for the ortholog from the same genus, *Pseudomonas aeruginosa* MurB

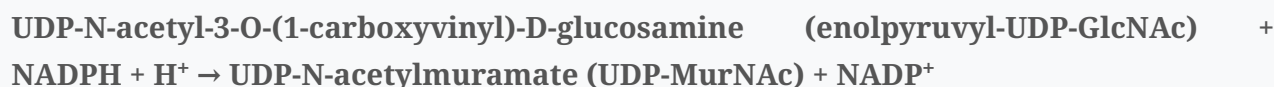
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providing a strong basis for direct inference. Function is therefore assigned by (i) unambiguous family/domain membership and (ii) homology to biochemically and structurally characterized MurB enzymes, most closely the *Pseudomonas* ortholog.

2. Primary Function: the reaction catalyzed

MurB catalyzes:



This is the reductive conversion of an enol ether to the D-lactyl ether that defines N-acetylmuramic acid. In the Mur pathway, MurA first transfers phosphoenolpyruvate onto UDP-GlcNAc to make the enolpyruvyl adduct; **MurB then reduces that enolpyruvyl group**, completing formation of UDP-MurNAc (Benson et al.,

P 8448160

Benson, Walsh & Hogle,).

P 8805513

Substrate specificity. The physiological substrates are highly specific: the sugar-nucleotide **enolpyruvyl-UDP-GlcNAc** as the reducible substrate and **NADPH** (not NADH) as the hydride donor. The enzyme contains one tightly, stoichiometrically bound **FAD** cofactor that is reduced and re-oxidized during each catalytic cycle (

P 8448160

The *E. coli* enzyme is a 38-kDa monomer purified to homogeneity with catalytically competent FAD

P 8448160

3. Catalytic Mechanism (flavin-mediated, two half-reactions)

MurB is a classic example of a flavoenzyme running a **ping-pong (two-half-reaction) mechanism** at a single active site:

1. **Reductive half-reaction (NADPH → FAD).** A hydride is transferred from NADPH — specifically the **4-pro-S hydrogen** — to enzyme-bound FAD, generating FADH₂ (Benson et al.,

P 8448160

In the *P. aeruginosa* ternary complex, the NADP⁺ nicotinamide ring stacks against the **si face of the FAD isoalloxazine**, an unusual geometry for hydride transfer (Chen et al.,

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2. **Oxidative half-reaction (FADH₂ → enolpyruvyl substrate).** FADH₂ reduces the enolpyruvyl group: a **hydride is delivered to the β-methyl** of the incipient 3-O-lactyl moiety while a **proton from solvent adds to the α-carbon**, producing the D-lactyl ether of UDP-MurNAc

P 8448160

Ser229 is the catalytic proton donor for this step — the S229A mutant suffers an ~10⁷-fold kinetic defect in the second half-reaction (Benson, Walsh & Hogle,

P 9020778

Single shared active site. After NADP⁺ departs, the sugar-nucleotide binds the *same* pocket used by the nicotinamide. In *P. aeruginosa* MurB, both substrates share a binding site located between two lobes of substrate-binding **domain III**, with the nicotinamide and enolpyruvyl moieties strikingly superimposable and both poised for hydride exchange with FAD; a **potassium ion** in the active site aids substrate orientation, and channel flexibility lets the non-reactive parts of the two substrates adopt different conformations

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Conformational gating. Substrate binding drives a large motion of **domain 3 (residues ~219–319)** that closes the substrate channel over the ligand; the Tyr190/Tyr254 stacking pair is disrupted, and Tyr190 swings over to hydrogen-bond the α-phosphate of the sugar-nucleotide (Benson, Walsh & Hogle,

P 8805513

NMR confirmed that NADP⁺ binding induces chemical-shift changes across both the sugar-nucleotide and FAD pockets (Constantine et al.,

P 9150408

Conserved catalytic residues. Site-directed mutagenesis in *S. aureus* MurB defined distinct roles: **Arg213** maintains the FAD electronic state, **Arg176** binds the enolpyruvyl-UDP-GlcNAc substrate, and **His259** contacts both the sugar-nucleotide and NADP⁺ (Nishida et al.,

P 16236703

These residues are expected to be conserved in the *P. putida* ortholog.

4. Subcellular Localization

MurB functions in the **cytoplasm**. Peptidoglycan biosynthesis is compartmentalized across cytoplasm, inner membrane, and periplasm; the **cytoplasmic phase forms the soluble nucleotide precursors** through the Mur enzymes, converting UDP-GlcNAc into UDP-MurNAc-pentapeptide (Amera et al.,

P 42128180

MurB is explicitly described as catalyzing "an essential cytoplasmic step in the synthesis of peptidoglycan" (Shapiro et al.,

P 22068704

It is a soluble, monomeric protein with no membrane-spanning segments (consistent with the FAD-bd_PCMH soluble fold), acting on freely diffusible sugar-nucleotide substrates before the pathway moves to the membrane (MraY/MurG produce lipid I/II).

5. Pathway Context and Biological Process

MurB occupies a fixed, non-redundant position in the linear **Mur (peptidoglycan precursor) pathway**:

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UDP-GlcNAc
  | MurA (enolpyruvyl transfer from PEP)
  ▼
enolpyruvyl-UDP-GlcNAc
  | MurB ← NADPH + FAD (THIS ENZYME, EC 1.3.1.98)
  ▼
UDP-MurNAc
  | MurC (add L-Ala)
  ▼
UDP-MurNAc-L-Ala → (MurD, MurE, MurF) → UDP-MurNAc-pentapeptide
  | MraY / MurG (membrane) → Lipid I / Lipid II
  ▼
peptidoglycan (periplasm)

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MurB's product UDP-MurNAc is the demonstrated substrate for the next enzyme MurC (UDP-MurNAc:L-alanine ligase), as shown by coupled enzymatic synthesis using MurA (MurZ) + MurB to make UDP-MurNAc that then supports the MurC ligase reaction (Raymond et al.,

[P 14659546](#)

see also MurC structure/function, Seo et al.,

[P 33950018](#)

The broader biological process is **biosynthesis of the murein sacculus**, required for cell growth, elongation, division, and osmotic protection (Dhar et al.,

[P 29185941](#)

6. Essentiality and Physiological Importance

- **Genetic essentiality.** In *Bacillus subtilis*, *murB* lies in the **dcw ("division and cell wall") cluster** immediately upstream of *divIB*; attempts to inactivate it were unsuccessful, and growth rate, morphology, and antibiotic resistance depend on the MurB expression level (Real & Henriques,

[P 16484183](#)

- **Conditional-lethal genetics.** In *Staphylococcus aureus*, temperature-sensitive *murB* point mutants **halt peptidoglycan synthesis** at the restrictive temperature with reduced reductase activity, confirming *murB* is essential for growth; five of six causal mutations map

to substrate-binding domain 3 (Matsuo et al.,

[P 12757953](#)

Nishida et al.,

[P 16236703](#)

- **Drug-target validation.** Because interrupting this step is lethal and the enzyme has no human counterpart, MurB is an actively pursued antibacterial target: multiple inhibitor classes (4-thiazolidinones, 3,5-dioxypyrazolidines/pyrazolidine-3,5-diones, tetrazoles) inhibit MurB and reduce peptidoglycan synthesis and bacterial growth (Yang et al.,

[P 16436710](#)

Gilbert et al.,

[P 17004716](#)

Hrast et al.,

[P 29363274](#)

and dedicated high-throughput assays have been developed (Shapiro et al.,

[P 22068704](#)

For *P. putida* (an environmental, non-pathogenic soil bacterium and biotechnology chassis), MurB performs the same housekeeping role in building and maintaining the Gram-negative cell wall; its essentiality is inferred from the universal conservation and demonstrated essentiality of MurB across bacteria.

7. Evidence Basis and Confidence

Claim	Evidence type	Confidence
Reaction catalyzed (enolpyruvyl-UDP-GlcNAc → UDP-MurNAc; NADPH-dependent)	Direct biochemistry (E. coli), family/domain annotation	High
FAD cofactor, monomer, 4-pro-S NADPH hydride	Direct biochemistry (P 8448160)	High
Ping-pong, single-site mechanism; NADP ⁺ /FAD geometry	X-ray structures incl. same-genus P. aeruginosa (P 23805286)	High
Ser229 proton donor; Arg/His/Tyr active-site network	Mutagenesis + structure (P 9020778 P 16236703 P 8805513)	High for family; inferred for PP_1904
Cytoplasmic localization	Pathway/biochemical consensus (P 22068704 P 42128180)	High
Pathway position (MurA→MurB→MurC)	Coupled enzymatic reconstitution (P 14659546)	High
Essentiality	Genetics in B. subtilis/S. aureus (P 16484183 P 12757953)	High for bacteria; inferred for P. putida

Key inference note: All mechanistic/structural data derive from orthologs (*E. coli*, *S. aureus*, *T. caldophilus*, and — closest — *P. aeruginosa*). Given >universal conservation of MurB catalytic residues and the availability of a *Pseudomonas* structure, transfer of annotation to *P. putida* PP_1904 is well supported.

8. Supported vs. Refuted Hypotheses

Supported: - MurB (Q88LM5) is the FAD-dependent UDP-N-acetylenolpyruvoylglucosamine reductase (EC 1.3.1.98). ✓ - It uses NADPH and a two-half-reaction flavin mechanism at a single active site. ✓ - It functions in the cytoplasm, upstream of MurC. ✓ - It is essential and a validated antibacterial target. ✓

Refuted / excluded: - Not a membrane transporter, structural/adaptor protein, or signaling molecule — it is a soluble metabolic oxidoreductase. ✓ - Does not use NADH preferentially (NADPH-specific). ✓ - Gene-symbol ambiguity: **not applicable** — literature is abundant and specific to this exact enzyme family; no conflicting "murB" gene in other organisms was encountered.

9. Limitations and Future Directions

- **No direct study of PP_1904** exists; function is assigned by homology (strongly supported by the *P. aeruginosa* structure). A recombinant enzyme assay or complementation of an *E. coli* *murB*(Ts) strain with PP_1904 would give direct confirmation.
- Kinetic parameters (kcat, Km for enolpyruvyl-UDP-GlcNAc and NADPH) specific to *P. putida* are unknown.
- Whether *P. putida* MurB participates in any species-specific protein–protein interactions with divisome/elongasome components (as reviewed generally in

P 27136593

has not been examined.

- Potential *P. putida*-specific roles in stress/solvent tolerance (a hallmark of this organism) via cell-wall remodeling remain unexplored.

References (PMIDs)

8448160 (Benson et al. 1993, mechanism); 8805513 (Benson, Walsh & Hogle 1996, substrate-free structure); 9020778 (Benson, Walsh & Hogle 1997, S229A); 9150408 (Constantine et al. 1997, NMR); 23805286 (Chen et al. 2013, *P. aeruginosa* MurB structure); 16236703 (Nishida et al. 2006, *S. aureus* active-site residues); 12757953 (Matsuo et al. 2003, *S. aureus* Ts mutants); 16484183 (Real & Henriques 2006, *B. subtilis* dcw/essentiality); 14659546 (Raymond et al. 2003, MurA+MurB→UDP-MurNAc→MurC); 33950018 (Seo et al. 2021, MurC); 22068704 (Shapiro et al. 2012, cytoplasmic/essential, assay); 29363274 (Hrast et al. 2018, inhibitors); 16436710 (Yang et al. 2006, inhibitors + *E. coli* structure); 17004716 (Gilbert et al. 2006, inhibitors); 42128180 (Amera et al. 2026, Mur enzyme review); 29185941 (Dhar et al. 2018, *E. coli*/*P. aeruginosa* cell-wall review); 12492849 (El Zoeiby et al. 2003, Mur enzyme review); 16934839 (Smith 2006, Mur ligases); 27136593 (Laddomada et al. 2016, cell-wall protein complexes).

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